

# **BHABHA UNIVERSITY, BHOPAL**

NH-12, Hoshangabad Road, Jatkhed, Bhopal-462026

## **POST GRADUATE DIPLOMA IN COMPUTER APPLICATIONS (PGDCA)**



### **DETAILED SYLLABUS (EFFECTIVE FROM JULY 2018)**

**Department of Computer Science & Application  
BHABHA UNIVERSITY, BHOPAL**

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**POST GRADUATE DIPLOMA IN COMPUTER APPLICATIONS**  
**(PGDCA)**

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**DETAILED SYLLABUS**

**SCHEME**

**SEMESTER -I**

| S.No.       | Paper Code | Paper Name              | Internal Evaluation | Theory | Practical | Total |
|-------------|------------|-------------------------|---------------------|--------|-----------|-------|
| 1           | PGDCA11    | Fundamental of computer | 30                  | 70     |           | 100   |
| 2           | PGDCA12    | PC Packages             | 30                  | 70     |           | 100   |
| 3           | PGDCA13    | Programming In C++      | 30                  | 70     |           | 100   |
| 4           | PGDCA14    | Operating System        | 30                  | 70     |           | 100   |
| PRACTICAL   |            |                         |                     |        |           |       |
| 5           | PGDCA15    | PC Packages             | 20                  |        | 30        | 50    |
| 6           | PGDCA16    | Programming In C++      | 20                  |        | 30        | 50    |
| TOTAL MARKS |            |                         | 160                 | 280    | 60        | 500   |

## **PAPER CODE - PGDCA11**

### **FUNDAMENTALS OF COMPUTER**

#### **UNIT- I**

Introduction to Computer and Information Technology: Brief history of development of computer & generations of computer, Computer system characteristics. Capabilities and limitations block diagram of computer. Types of computer-Analog, Hybrid, digital, micro, mini, mainframe, super computer. Personal computer, types of PCs desktop, laptop, notebook, palmtop etc. Number system, Number system of computers binary, octal, hexadecimal, representation & their conversion, Coding system ASCII, BCD, EDCDIC etc.

#### **UNIT- II**

INPUT/OUTPUT devices: keyboard, mouse, monitor, trackball, joystick, digitizing table, scanners, digital cameras, MICR, OCR, OMR, Bar-code reader, Voice recognition, light pen, touch screen, devices, printer, plotter.

#### **UNIT- III**

Storage device: Data storage and retrieval methods-sequential, direct and index sequential-various storage devices-magnetic tape, magnetic disks, cartridge tape, data drives hard disk drives, floppy disks, optical disks-CD, VCD, CDR, CDRW, DVD.

#### **UNIT- IV**

Computer software: types of software, system software, application software, operating system, utility program, assemblers, compilers and interpreter. Operating system functions, Types batch, single user, multi user, multiprogramming, multiprocessing, Programming languages, machine, assembly, high level, 4GL, their merits and demerits. Computer virus – types of virus, virus detection & prevention virus on network.

#### **UNIT- V**

Data Communication & networks: analog and digital signals, modulations, amplitude modular (am), frequency modulation (fm), phase modulation (pm), communication process, direction of transmission flow, simplex, half duplex, full duplex. Types of network LAN, WAN, MAN etc, Topologies of LAN ring, bus star, mesh and tree topologies, communication protocols TCP/IP protocol suit. Communication channels media twisted, coaxial fiber optic.

#### **Text & Reference Books:**

01. Computer fundamentals, P.K. Sinha, BPB
02. Computer today by S.K. Basandra Galgotia Publications.
03. Fundamentals of information by Axexos Leon & Mathews Leon, Vikas Publishing House, New Delhi New Delhi

## **PAPER CODE - PGDCA12**

### **PC PACKAGES**

#### **UNIT- I**

Fundamental of DOS & Windows: Fundamental of DOS booting process, internal and external commands, creating and executing batch files and directories creating text files. Introduction to windows features, various versions of windows, origin of windows parts of windows screen types and anatomy of windows using.

#### **UNIT- II**

Introduction to word processing (MS-word): Advantages of word processing, editing a file using paragraphs, bullets, indentation, ect. Formatting features, printing the documents, it includes paper-size, margins, header and footer, page no., using macros. Advance word processing, header and footers. Finding text, mail merge and other application, mathematical calculations, table handing.

#### **UNIT- III**

Introduction to spread sheet (MS-Excel): Definition and advantages of electronic worksheet, working of spread sheet, range and related operations. Setting saving and retrieving work sheet file, inserting deleting coping & moving of data cells, inserting and deleting rows & columns, protecting cell printing a worksheet, erasing a worksheet.

#### **UNIT- IV**

Introduction to Power point (MS- Power point): Creating a presentation, inserting/deleting slides, different slide views, editing slides,. Slide transition & editing special effects inserting sound, picture, chart, organization chart.

#### **Text & Reference Books:**

1. Comdex Computer Course Kit (Windows 7 with office 2010), Gupta vikas, Dreamtech
2. Mastering MS Office 2000, Professional Edition by Courter, BPB Publication.
3. MS Office 2000 Training Guide by Maria, BPB Publication.
4. PC Software, Ravi Taxalli, BPB
5. Computer Fundamental by P.K. Sinha

## **PAPER CODE - PGDCA13**

### **PROGRAMMING IN C++**

#### **UNIT-1**

Fundamentals of C: I/O statements, Assignment Statements, Constants, Variables, Operators and Expressions, Standards and Formatted statements, Keywords, Data Types and Identifiers. Control Structures: Introduction, Decision making with if – statement, if-else and Nested if, while and do-while, for loop. Jump statements: break, continue, goto, switch Statement.

#### **UNIT-2**

Structure and Union: Declaration of structure, Accessing structure members, Structure Initialization Overview of OOP: Classes & Functions, Scope Resolution Operator, Private and Public Member Functions, Nesting of Member Functions. Creating Objects, Accessing class data members, Accessing member functions.

#### **UNIT-3**

Arrays: Introduction to Arrays, Array Declaration, Single and Multidimensional Array, Memory Representation, Matrices, Strings, String handling functions.

#### **UNIT-4**

Inheritance - Extending Classes Concept of inheritance, Base class, Derived class, Defining derived classes. Types of Inheritance Polymorphism: Definition, early Binding, Polymorphism with pointers, Virtual Functions, late binding, pure virtual functions.

#### **UNIT-5**

Pointers: Introduction to Pointers, Address operator and pointers, Declaring and Initializing pointers, Assignment through pointers. File Management in C++: Defining & opening a file, closing a file, I/O operations on file, error handling during I/O operations, Random Access Files.

#### **Text & Reference Books:**

1. Balagurusamy, “Programming in C”, 5<sup>th</sup> Edition, Tata McGraw-Hill Education, 2007
2. Yashavant Kanetkar, “Let us C”, 10<sup>th</sup> Edition, BPB publication, 2010
3. Balagurusamy, “Object Oriented Programming with C++”, 3<sup>rd</sup> Edition, Tata McGraw-Hill Education, 2006

## **PAPER CODE - PGDCA14**

### **Operating System**

#### **UNIT-1**

**INTRODUCTION:** - Machine Hardware Traps and Interrupts, Multimode Execution, operating System Structure, Operating System Types, the Boot Process. **PROCESS MANAGEMENT:** - Process Scheduling, Process State, Scheduling Criteria, Scheduling Algorithms First-Come First-Served, Round Robin, and Priority.

#### **UNIT-2**

**INTERPROCESS COMMUNICATION AND SYNCHRONIZATION:** - Interprocess Communication, Process Synchronization Critical Section, Interrupt Disabling, Test and Set Instruction, Write a Program Instruction, Wait and Signal, Semaphores.

#### **UNIT-3**

**Deadlock** (Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock). **MEMORY MANAGEMENT:** - Paging, Segmentation, Segmentation with Paging, Page and Segment Tables (Associative Memory, Inverted Page Table).

#### **UNIT-4**

**VIRTUAL MEMORY:** - Demand Paging Locality of Reference, Page Locking, Page Size, Page Replacement Algorithms, Algorithm Performance, Allocation Policies, Working Set, **FILE SYSTEM MANAGEMENT:** - Directories and Names Partitions, Per-Process Root Directory, Directory Structure, Directory Entries, Types of File System Objects, File System Functions, Information Types, File System Architecture ,Access Methods, Access Control.

#### **UNIT-5**

**DEVICE MANAGEMENT:** - Hardware I/O Organization (I/O Control, Port and Memory-Mapped I/O, Module Registers, Busy Wait I/O, Polled I/O, Interrupt I/O, Direct Memory Access (DMA)), **SECURITY:** - Authentication (Passwords, Physical Authentication), Prevention, Detection, Correction, Identification, Threat Categories, Program Threats.

#### **Text & Reference Books:**

1. William Stalling, "Operating System Internals and Design Principle", edition 6<sup>th</sup>, Pearson Education India, 2009.
2. Peter bears Galvin, "Operating System Principle", Edition 7<sup>th</sup>, Wiley India, 2009
3. J. Harris, "Operating System SCHAUM'S OUTLINE", Special Indian edition,